Beyond the First Visit

CMAT attendance and mathematics outcomes. Paper 2 - provisional scientific synthesis, 26 October 2026. For research faculty and CMAT administration.

Source: Consolidation plan

Principal message

The strongest evidence distinguishes zero recorded visits from initial use. Adjusted outcome differences are sizeable, while positive-frequency increments are less precise and model-dependent. No causal benefits are identified.

Source: Paper 2.1 Wald/Holm and 10_benchmark

Why mathematics support matters

Learning centres supplement instruction amid different levels of preparation. Use is self-selected; non-engagement may reflect access barriers. Experiments and observational studies provide different types of evidence.

Source: Lawson et al. 2020; Mullen et al. 2024; Pugatch & Wilson 2018

CMAT and the optional PPA1 activity

PPA1 provides one optional three-CMAT-visit integration activity. The study does not observe individual intentions, PPA eligibility or credit use. Three visits motivate an exploratory grouping, not a discontinuity design.

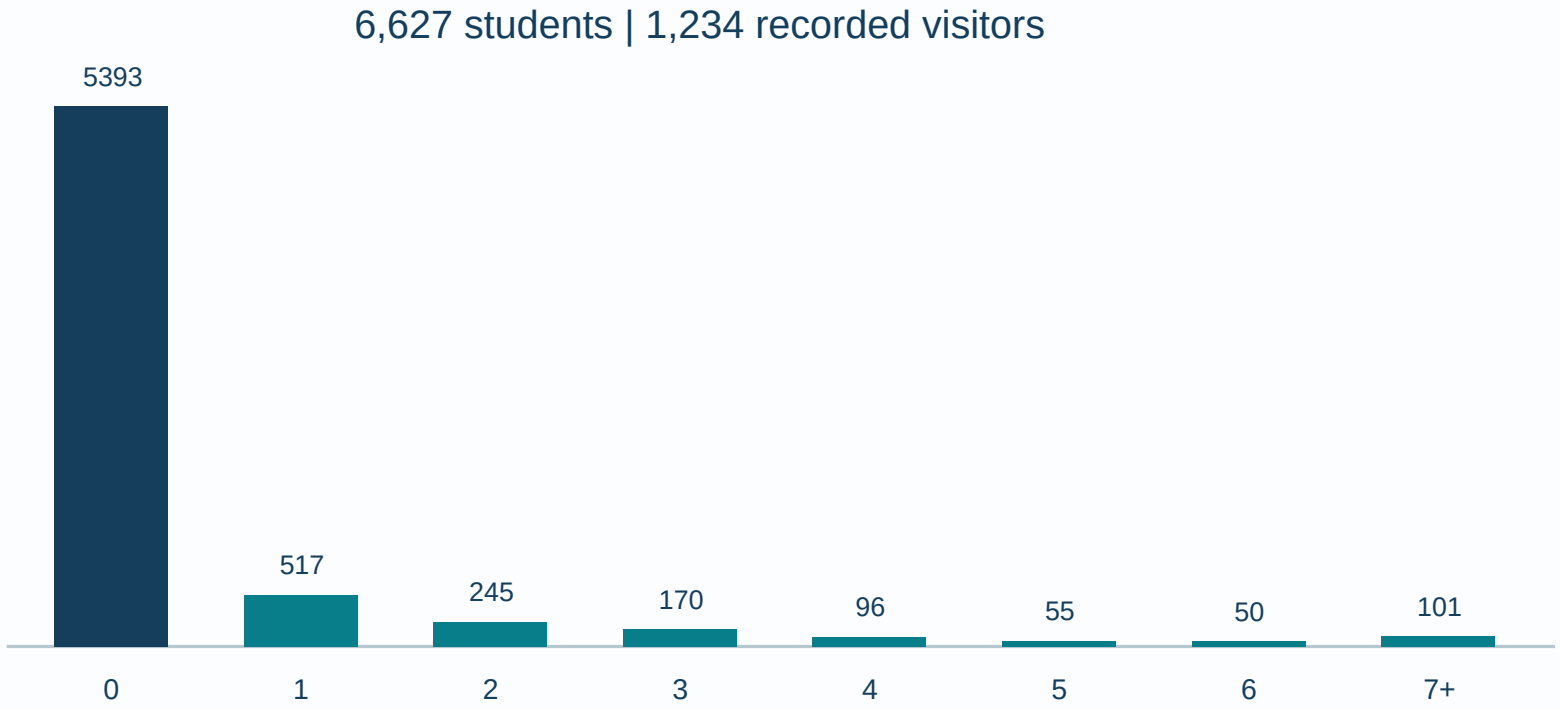
Source: Consolidation plan, 2026-10-26

Cohort and empirical design

First eligible MU attempt with CMAT coverage, context fixed effects, degree adjustment and professor-period clustered uncertainty. Primary outcomes: Z_GRADE_PRIMARY, PASS; numeric-only and administrative sensitivities differ in sample and target.

Source: Paper 2.1 specifications; consolidation plan

Visit distribution and precision

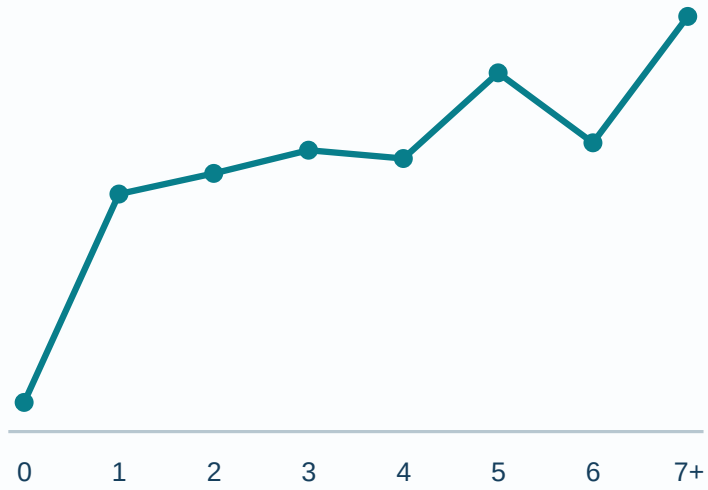


N=6,627; 190 professor-period contexts; 5,393 with zero visits and 1,234 with positive recorded visits. Sparse high-frequency support constrains fine-grained contrasts.

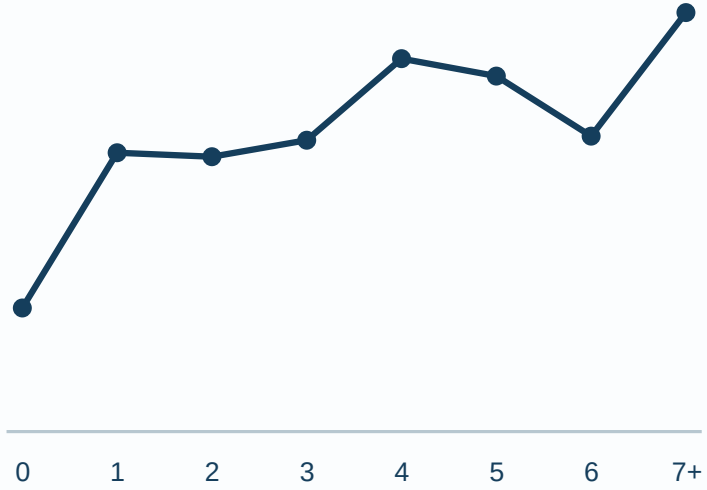
Source: 01_cohort_flow.csv; 10n_zero_inclusive_7plus_descriptives.csv

Raw grades and passing rates

Raw standardized grade (SD)



Raw PASS rate (%)



These are unadjusted between-student means, not longitudinal growth or experimental effects. The exact-frequency series is not strictly monotone.

Source: 10n_zero_inclusive_7plus_descriptives.csv

Any attendance vs zero

Adjusted 1+ minus 0: +0.361 grade SD, CI [0.302,0.421], and +15.29 PASS percentage points, CI [12.44,18.15]. The 1+ pool is not exactly one visit.

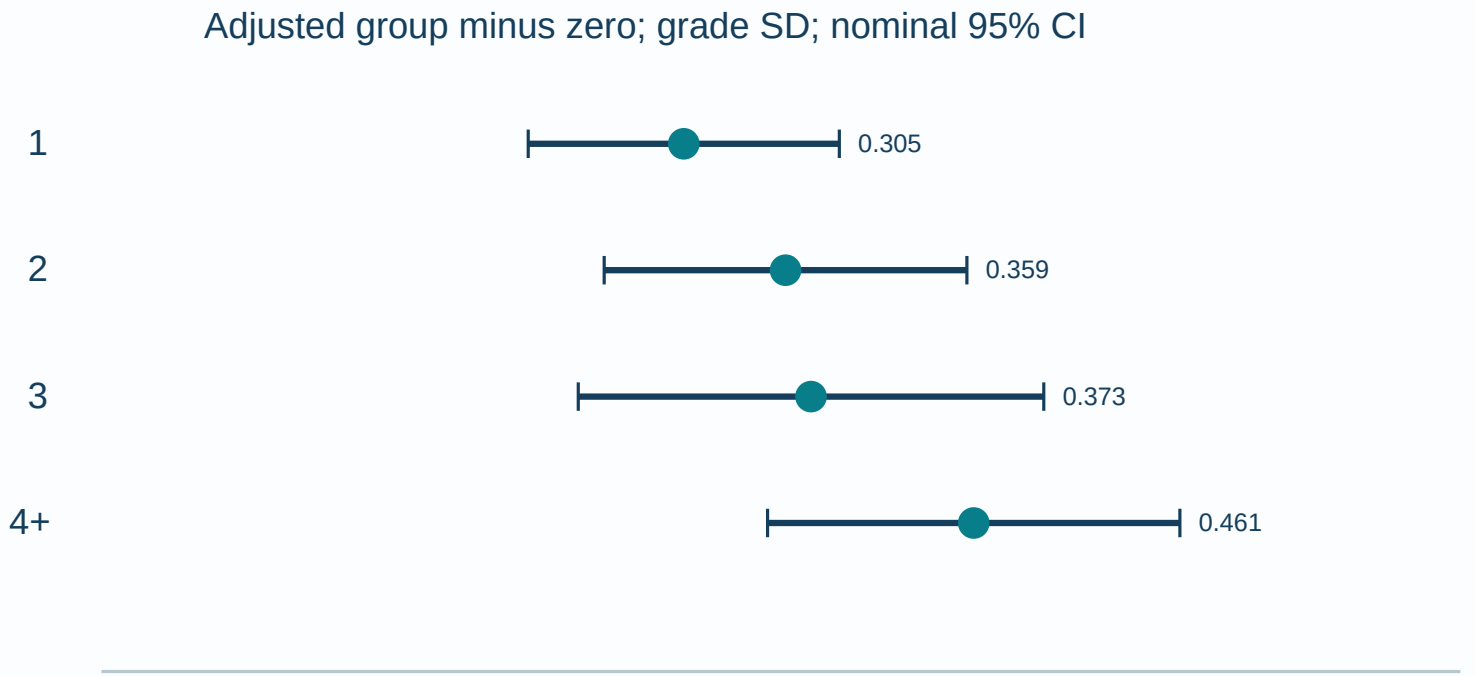
Source: 10_benchmark_0_vs_1plus.csv

Wald and Holm inference

The original 0 | 1 | 2 | 3 | 4+ design has ten pairwise contrasts per outcome. Holm adjustment is within this specified family; the eight-category adjacent family and new PPA grouping are separate.

Source: Paper 2.1 tables 33 and 41

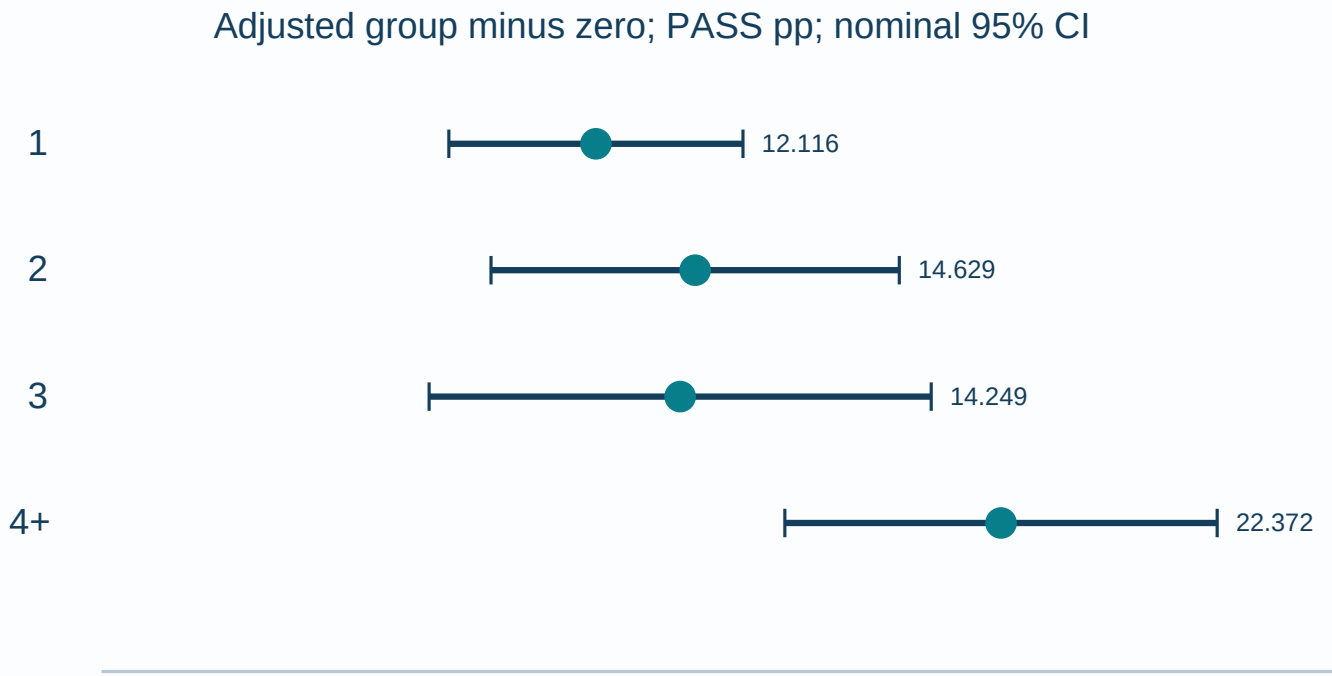
Adjusted standardized-grade contrasts



Each positive group differs from zero after ten-pair Holm correction. No positive-versus-positive grade contrast survives the ten-pair family.

Source: 33_exact_visit_groups_fe_pairwise.csv

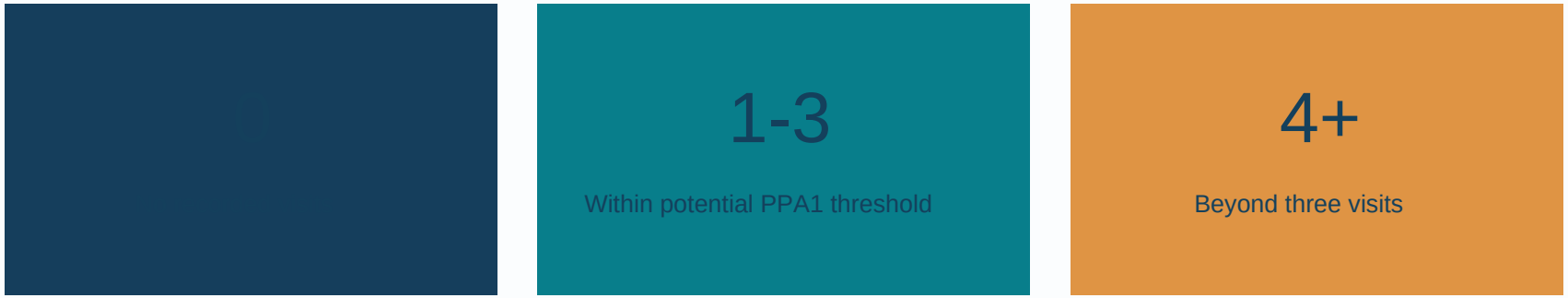
Adjusted PASS contrasts



Every positive category differs from zero after Holm; 1 versus 4+ also survives (Holm p~0.011). Other positive-pair differences do not.

Source: 41_exact_visit_groups_pass_fe_pairwise.csv

PPA grouping: not yet validated



Conceptual categories - PPA analysis pending validation

Primary 0 | 1-3 | 4+; four-group sensitivity 0 | 1-3 | 4 | 5+. Pending validated adjusted estimates. The figure is a grouping diagram, not a plotted result.

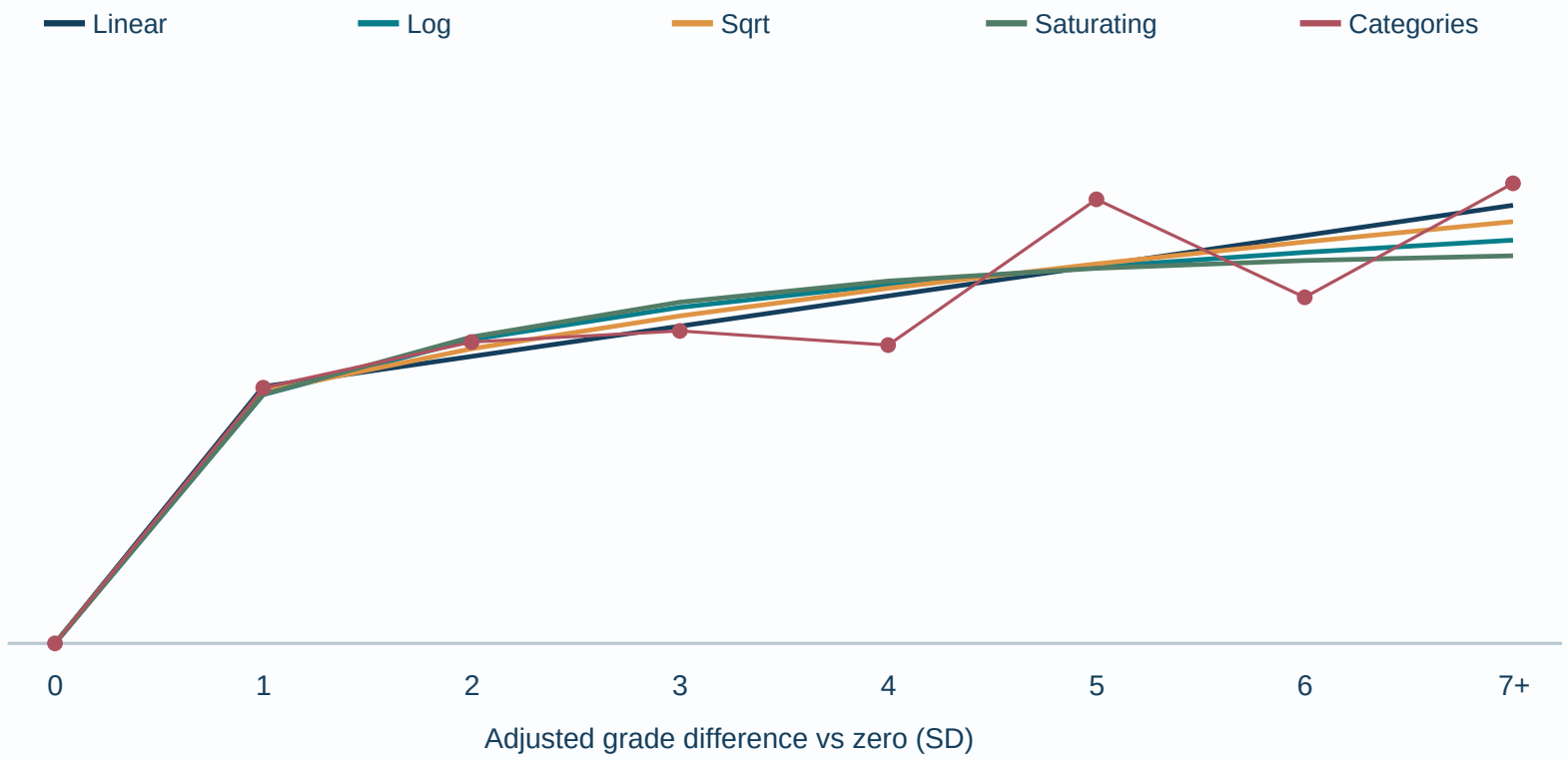
Source: PAPER2_CONSOLIDATION_PLAN.md

Two-part first/additional visits

Adjusted working model: Z first-visit +0.307 SD, additional-visit slope +0.036 SD; PASS first +11.92 pp, extra +2.23 pp. The common post-first slope is imposed and top-coded at 7+.

Source: two_part_attendance_z_pass.csv

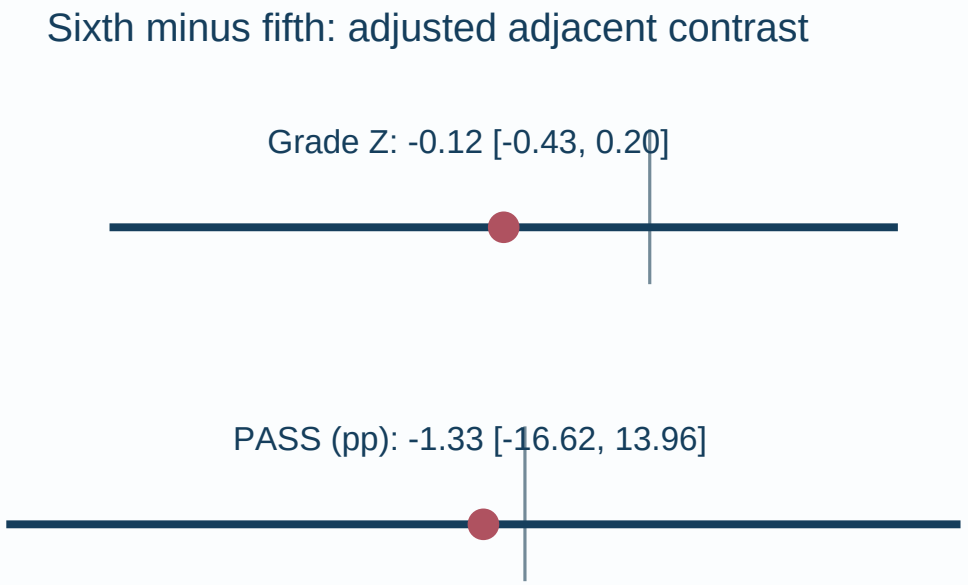
Five fitted curve families



Linear, log, square-root and saturating fits impose different increment shapes; categorical effects do not. Slightly better in-sample AIC or grouped validation does not prove causal diminishing returns.

Source: attendance_curves/adjusted_curves_cluster_ci.csv

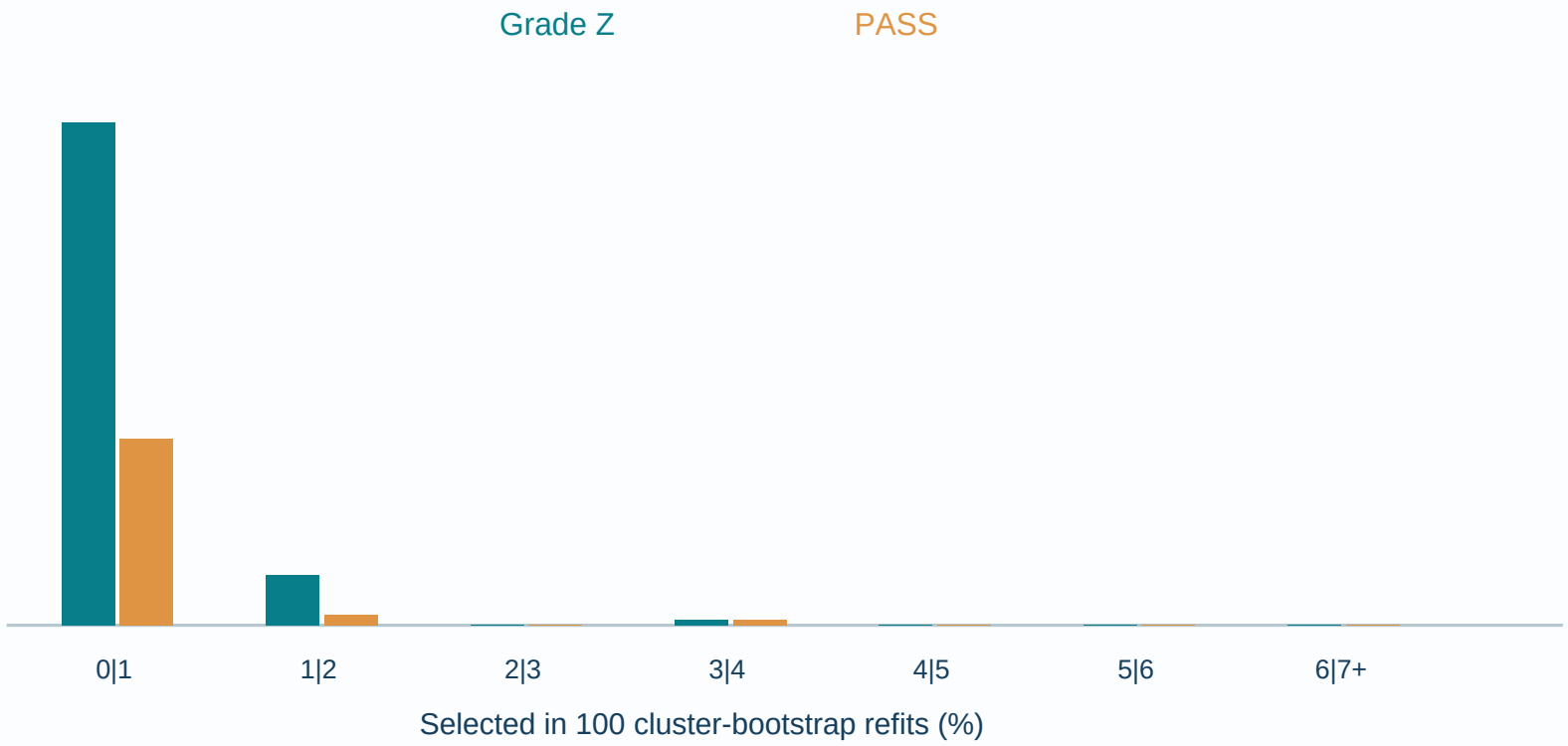
Five-to-six dip: broad uncertainty



Sixth minus fifth adjusted contrast: grade -0.117 SD, CI [-0.433,+0.199]; PASS -1.33 pp, CI [-16.62,+13.96]. Neither survives seven-adjacent Holm. N5=55, N6=50.

Source: attendance_curves/adjacent_differences_holm.csv

Fused-lasso boundary stability



Grade discovery selects [0] | [1-7+]; 0|1 boundary appears in 100/100 cluster bootstrap refits. PASS one-SE-style selection fuses all counts, although its 0|1 boundary appears 37/100 times.

Source: fused 7plus blocks and boundary_stability CSVs

Internally held-out grade comparison

Reserved n=1,952 in 57 contexts: discovered Z group 1+ minus zero +0.264 SD, CI [0.150,0.377]. PASS has no selected boundary, so no comparable selected Wald contrast.

Not an external or prospectively untouched replication.

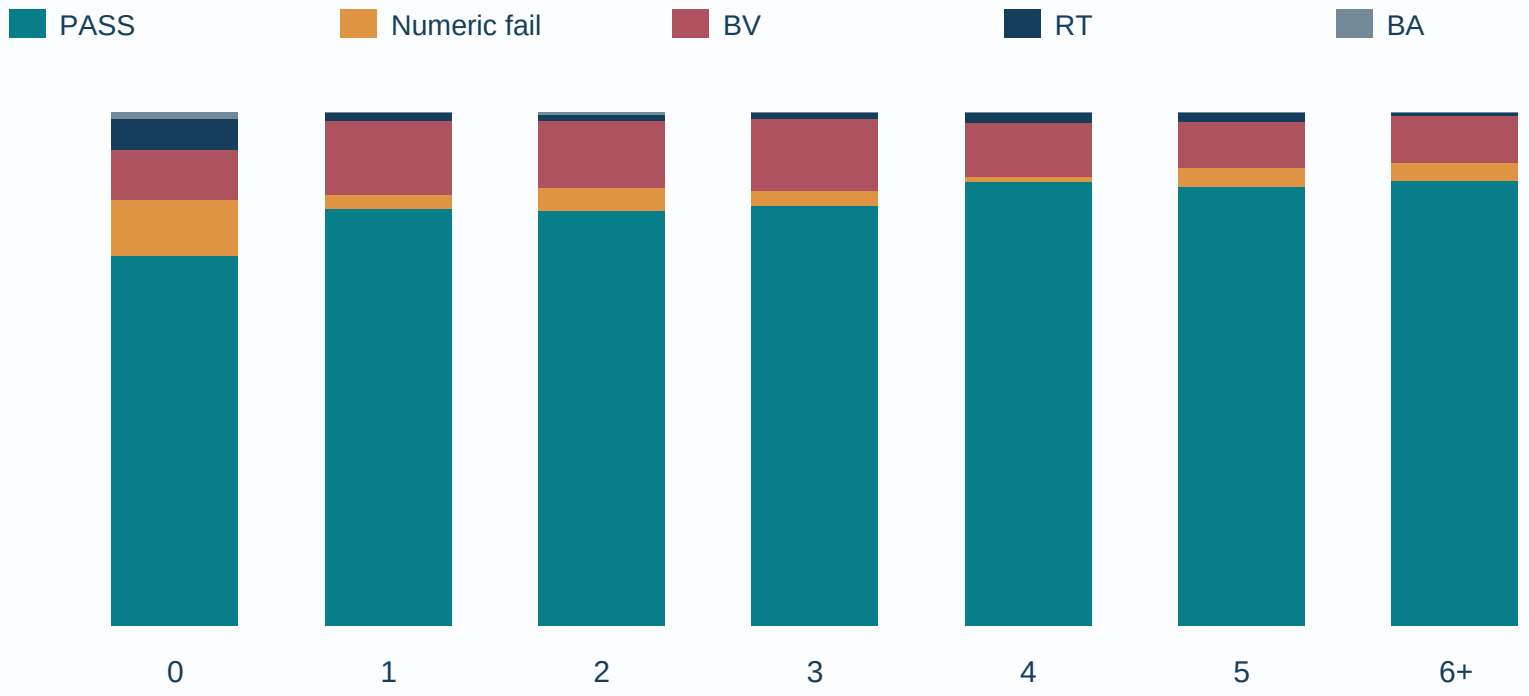
Source: 7plus_z_grade_primary_holdout_wald.csv

Mixtures and skew-normal alternatives

Historical EM Gaussian mixtures and skew-normal fits test distributional adequacy under different outcome encodings. A better K=2 likelihood does not identify two student types or even two visible modes. Prospective modality test remains pending.

Source: Distributional branch results/paper21/tables/37-52

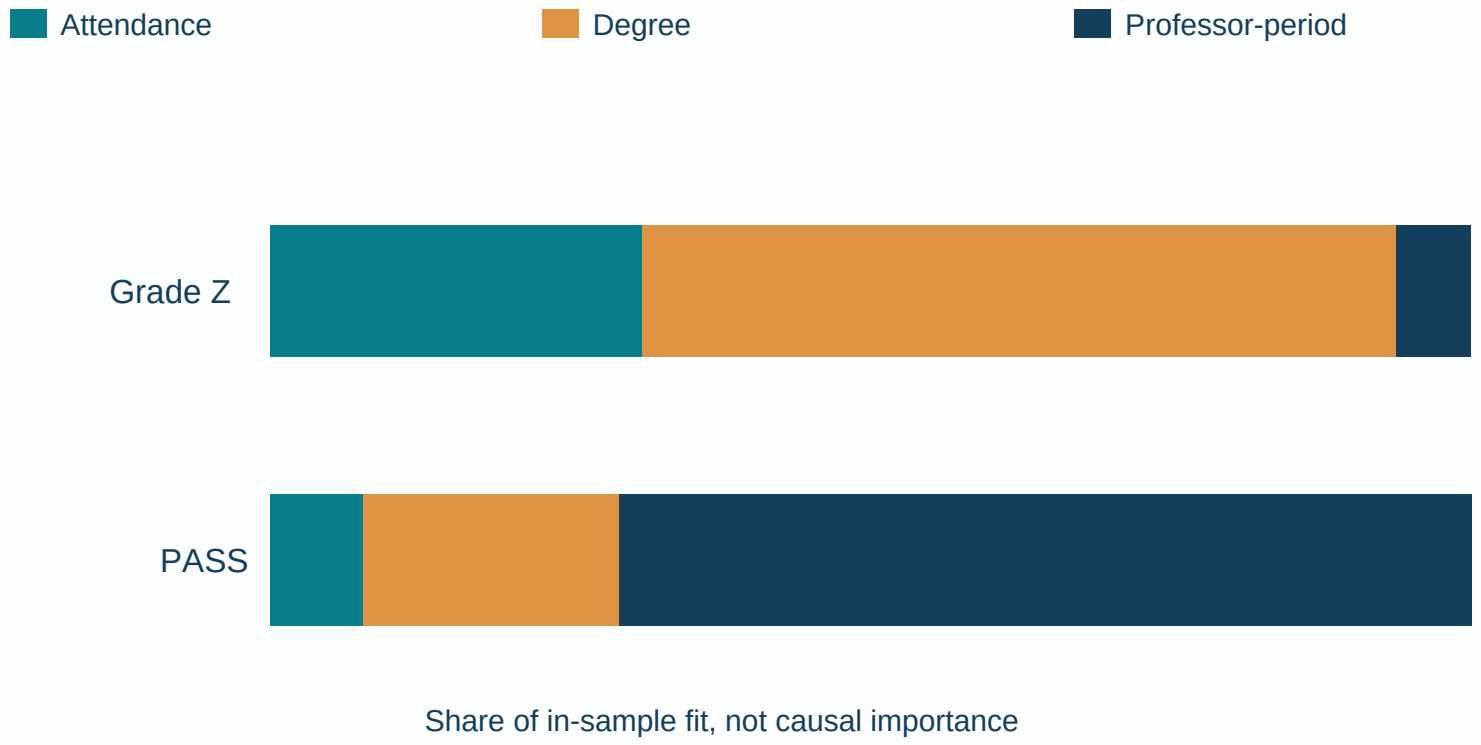
Observed outcomes and administration



PASS, numeric fail, BV, RT and BA are different observed states. Numeric-only grades exclude administrative outcomes; primary Z applies defined imputation. Combining these populations changes the estimand.

Source: 10d_zero_inclusive_exact_administrative_composition.csv

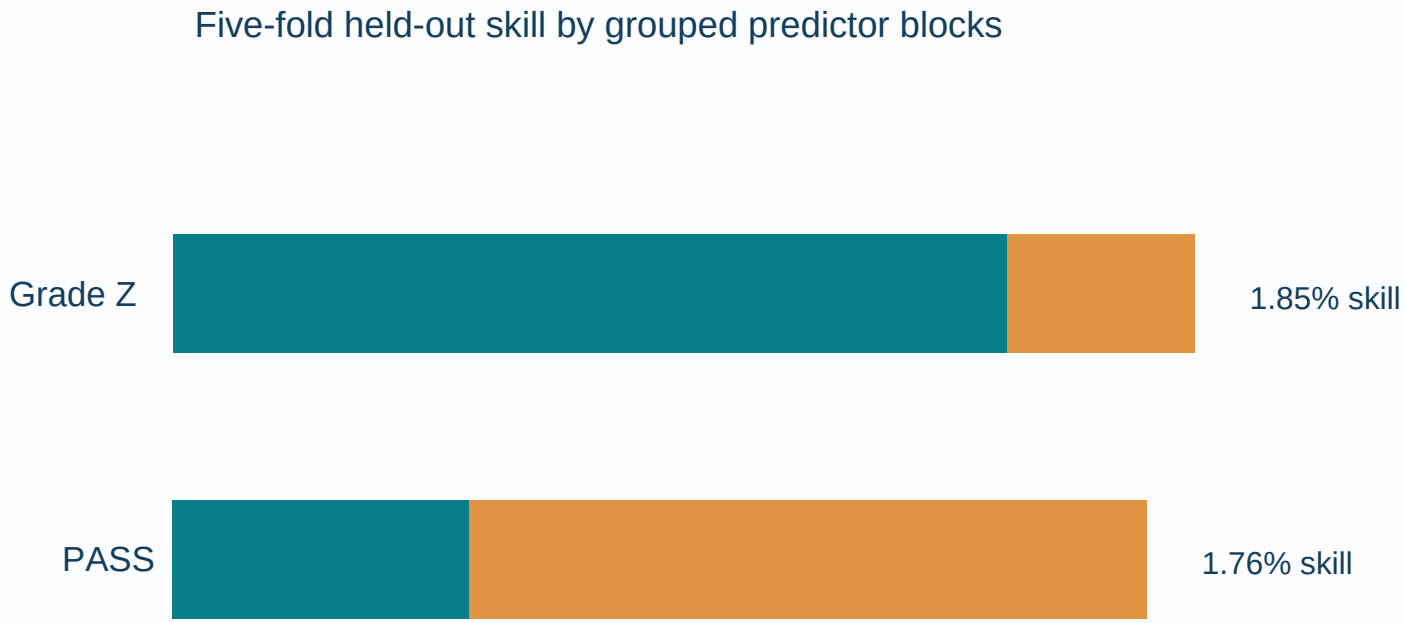
In-sample Shapley fit attribution



Model D: attendance accounts for approximately 31% of grade R2 and 8% of PASS R2. Degree dominates grade fit; professor-period dominates PASS fit. These are not causal contributions.

Source: shapley_importance/in_sample_shapley_12_models.csv

Out-of-context predictive skill



Five-fold professor-period-held-out analysis: grade R2 skill 1.85%, PASS Brier skill 1.76%. Test-context FE indicators are unavailable. These are retrospective predictions, not actionable start-of-term warnings.

Source: shapley_importance/out_of_context_shapley.csv

Synthesis and next study

Supported: large adjusted zero/initial separation, uncertain exact increments, a stable grade-selection cut and heterogeneous outcome composition. Next: validate PPA groups, audit cohort encoding, record prior readiness and visit timing, and prospectively test encouragement.

Source: Integrated synthesis, provisional, 2026-10-26

Appendix A - evidence hierarchy

Primary: historical FE/Wald/Holm contrasts. Exploratory: PPA grouping pending; two-part and attendance curves; fused lasso; mixture diagnostics; Shapley and retrospective skill. These are different estimands, not independent confirmatory replications.

Source: Consolidation plan

Appendix B - cohort numbers

Eligible first MU attempts across all historical periods: 7,777; covered analysis: 6,627. Exact n: 0=5393, 1=517, 2=245, 3=170, 4=96, 5=55, 6=50, 7+=101. 190 professor-period contexts.

Source: 01_cohort_flow.csv; 10n_zero_inclusive_7plus_descriptives.csv

Appendix C - multiplicity

Wald tests compare context-adjusted mu(i)-mu(j) using clustered standard errors.

Original historical ten-pair family, separately declared seven-adjacent family and two-term piecewise family require distinct Holm corrections.

Source: Paper 2.1 tables 33 and 41; Holm (1979)

Appendix D - piecewise formula

Y=alpha_context + gamma_degree + beta_0 ANY_VISIT + beta_plus EXTRA + error.

ANY_VISIT=1(K>=1); EXTRA=max(min(K,7)-1,0). Z betas 0.30656/0.03609; PASS betas 0.11923/0.02225.

Source: two_part_attendance_z_pass.csv

Appendix E - shape-selection sensitivity

Z AIC: linear 18753.43, log 18753.99, sqrt 18753.62, saturating 18754.40, categorical 18762.27. Out-of-context Z R2 skill across shapes 0.01630-0.01692; differences are small.

Source: model_comparison_in_sample.csv; group_heldout_shape_cv.csv

Appendix F - fused protocol

Discovery n=4,675 / 133 contexts, reserve n=1,952 / 57 contexts; 100/100 bootstrap retunings successful. Z selected 0|1 in 100%. PASS all-fused under one-SE-style heuristic but 128/128 by exhaustive discovery BIC.

Source: 2026-10-07_fused_lasso_100_bootstrap_results.md

Appendix G - distributions

Numeric complete-case BIC prefers GMM K=2 vs single skew-normal in groups 0,1,2; skew-normal K=1 in 3,4,5,6+. Held-out log predictive density also varies. Do not infer underlying social subgroups.

Source: 47_complete_case_skewnormal_vs_gmm.csv; 50_complete_case_skewnormal_vs_gmm_cross_validation.csv

Appendix H - provenance and readiness

Paper 2.1 1546c594; fused 29810a86; mixture b1cace12; Shapley 5915976f. Claims matrix, integrated manuscript, validated PPA grouping pending at deck authoring. No submission-readiness claim.

Source: GitHub source tips verified 2026-10-26